Testbench Code

```
`timescale 1ns / 1ps
module tb_traffic_light_controller;
reg clk;
reg reset;
reg pedestrian;
wire [2:0] highway;
wire [2:0] side_road;
wire ped_walk;
// DUT
traffic_light_controller DUT
(
 .clk(clk),
 .reset(reset),
 .pedestrian(pedestrian),
 .highway(highway),
 .side_road(side_road),
 .ped_walk(ped_walk)
);
```

```
// Clock
always #5 clk = ~clk;
// Test
initial
begin
 clk = 0;
 reset = 1;
 pedestrian = 0;
 #20;
 reset = 0;
```

```
#80;
 pedestrian = 1;
 #10;
 pedestrian = 0;
 #150;
 $finish;
end
// Monitor
initial
begin
$monitor("Time=%0t | Ped=%b | Highway=%b | Side=%b | Walk=%b",
 $time,
 pedestrian,
 highway,
 side_road,
 ped_walk);
end
endmodule
```